Jonas Elias El Gammal

Cosmologist | Scientific machine-learning researcher | Scientific software developer

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PROFILE

Postdoctoral researcher and technology founder working across cosmology, Bayesian inference, scientific machine learning, and software. I develop validated computational methods and lead the translation of data, automation, and AI into operational systems.

PROFESSIONAL AND RESEARCH EXPERIENCE

Co-founder, Managing Director & CTO - Bright Interface GmbH 2026-present
- Lead product and technology strategy for a healthcare-software platform combining data integration, workflow automation, operational analytics, and applied AI.

Postdoctoral Researcher - KBFI / NICPB, Tallinn 2026-present
- KOHTO grant recipient; research in cosmology, gravitational-wave inference, scientific ML, and computational methods.

Postdoctoral Researcher - University of Insubria 2025-2026
- Developed GPU-accelerated and differentiable pulsar-timing-array inference with JAX in the EPTA collaboration.

Visiting Researcher - CERN 2024
- Developed a LISA analysis pipeline for scalar-induced gravitational waves.

EDUCATION

PhD in Physics - University of Stavanger 2021-2025
Developed data-analysis methods for LISA, including Gaussian-process-accelerated Bayesian inference and reconstruction of primordial curvature perturbations from scalar-induced gravitational waves.

Erasmus Exchange - Grenoble INP, PHELMA 2019-2020
Specialisation in nuclear engineering.

MSc in Physics - RWTH Aachen University 2018-2020
Grade 1.3 (approximately 3.7/4.0 GPA). Thesis: *Accelerating Bayesian Inference of Expensive Likelihoods with Gaussian Processes*.

BSc in Physics - RWTH Aachen University 2015-2018
Grade 1.7 (approximately 3.3/4.0 GPA). Focus on astroparticle physics and cosmology.

TEACHING

Numerical Methods, University of Stavanger - full bachelor-level course (2023, 2024): Python, numerical differentiation and integration, ODEs, and PDEs.

Electromagnetism and Special Relativity, University of Stavanger - full bachelor-level course (2023).

Physics and data-processing tutorials, RWTH Aachen University (2017-2019).

METHODS AND TECHNOLOGY

Research: Bayesian inference, Gaussian processes, active learning, uncertainty quantification, Monte Carlo methods, numerical optimisation, stochastic simulation, time-series analysis, cosmology, and gravitational-wave data analysis.

AI systems: LLM agents, tool use, agent loops, multi-agent workflows, typed planning, structured outputs, validation, repair, tracing, evaluation, and reproducible experimentation.

Software: Python, JAX, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow, C++, Git, Docker, Linux, SLURM, HPC/GPU computing, CI/CD, testing, documentation, and package development.

LANGUAGES

German - native | **English** - near-native | **French** - B2/C1

PUBLICATIONS

Are PTA measurements sensitive to gravitational wave non-Gaussianities?
C. Cecchini, J. El Gammal, G. Franciolini, M. Pieroni - 2026 - accepted for publication in Physical Review Letters; arXiv:2605.05157

Accelerating LISA inference with Gaussian processes
J. El Gammal, R. Buscicchio, G. Nardini, J. Torrado - 2025 - Physical Review D 112, 063010

Reconstructing primordial curvature perturbations via scalar-induced gravitational waves with LISA
J. El Gammal et al. (LISA Cosmology Working Group) - 2025 - JCAP 05, 062

Parallelized Acquisition for Active Learning using Monte Carlo Sampling
J. Torrado, N. Schoeneberg, J. El Gammal - 2023 - arXiv:2305.19267

Fast and robust Bayesian inference using Gaussian processes with GPry
J. El Gammal, N. Schoeneberg, J. Torrado, C. Fidler - 2023 - JCAP 10, 021

CURRENT RESEARCH AND DEVELOPMENT

- High-dimensional GP surrogate inference using gradient-based NUTS acquisition, with emphasis on numerical robustness, reproducible benchmarking, and scalable active learning.
- GPU-accelerated JAX likelihoods and reversible-jump inference for continuous gravitational-wave searches with pulsar timing arrays.
- PROSECCO: differentiable surrogate refitting of physical stochastic-background models against model-independent LISA reconstructions (in development).

OPEN-SOURCE SOFTWARE

GPry - Gaussian-process surrogate modelling and active learning for Bayesian inference with expensive likelihoods.

SIGWAY - JAX framework for scalar-induced gravitational-wave simulation and inference.

discoverysamplers - Consistent interfaces connecting Discovery PTA models to multiple Bayesian inference backends.

fastropop - JAX-backed simulation of supermassive black-hole-binary populations and their nanohertz signals.

Graphsmith - Typed graph planning with deterministic compilation, validation, local repair, and traceable execution.